Regression Diagnostics and Model Evaluation of Housing Prices

College of Professional Studies, Northeastern University

ALY 6015 | Module 01 | Assignment 01

2026-01-13

Isuri Nawodya Herath S W M M G
NUID – 002021794

Table of Contents

INTRODUCTION 3
ANALYSIS 3
Exploratory Data Analysis 3
Correlation Analysis 6
Scatterplot Comparison 7
Initial Regression Model (Model 1) 9
Diagnostic Evaluation 10
Multicollinearity Assessment 11
Outlier Analysis 12
Improved Model (Model 2) 13
All Subsets Regression 14
Model Comparison 2
Conclusion 3
REFERENCES 5
APPENDIX A : R CODE 6

INTRODUCTION

Regression analysis serves as a foundational tool in predictive analytics, enabling researchers and practitioners to understand relationships between variables and make informed predictions. This report presents a comprehensive regression analysis of the Ames Housing dataset, which contains detailed information about residential property sales in Ames, Iowa, from 2006 to 2010. The primary objective of this analysis is to develop, evaluate, and refine multiple linear regression models capable of accurately predicting housing sale prices based on various property characteristics.

The analysis emphasizes the identification and correction of common regression issues, including non-linearity, multicollinearity, outliers, and influential observations. Diagnostic plots and statistical measures are used to assess model assumptions and guide refinement. In addition, an automated all subsets regression approach is applied to identify a statistically optimal model for comparison.

This report follows a structured analytical approach, beginning with exploratory data analysis and correlation assessment, progressing through model fitting and diagnostic evaluation, and concluding with model refinement and comparison. The methodology employs R programming language with specialized packages including tidyverse for data manipulation, car for diagnostic testing, corrplot for correlation visualization, and leaps for automated model selection.

ANALYSIS

Exploratory Data Analysis

The initial exploration of the Ames Housing dataset revealed important characteristics about the data structure and distribution patterns. The dataset contains 2,930 residential property records with 82 variables, including 23 nominal, 23 ordinal, 14 discrete, and 20 continuous variables. Histograms were generated for key numeric variables to understand their distributions, as shown in Figure 1.

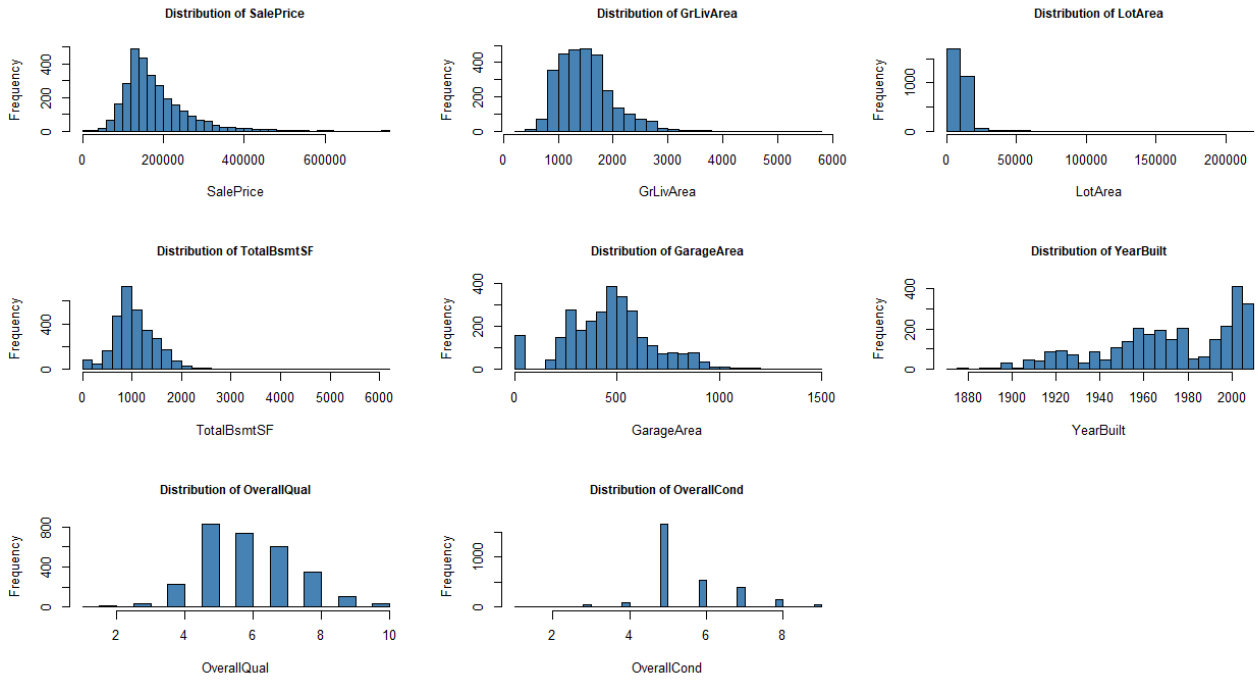

Figure 1 - Distribution of Key Numeric

The histograms reveal several important distributional characteristics. SalePrice exhibits a right-skewed distribution with most properties selling between $100,000 and $250,000, with a long tail extending toward higher values exceeding $600,000.

GrLivArea (above-grade living area) shows an approximately normal distribution centered around 1,500 square feet. LotArea displays severe right-skewness with extreme outliers exceeding 150,000 square feet. TotalBsmtSF and GarageArea both show right-skewed distributions with some zero values representing properties without basements or garages.

YearBuilt reveals a bimodal pattern with peaks around 1960 and 2000-2010, reflecting construction booms in those periods. OverallQual follows an approximately normal distribution centered at ratings of 5-6, while OverallCond shows a strong concentration at rating 5.

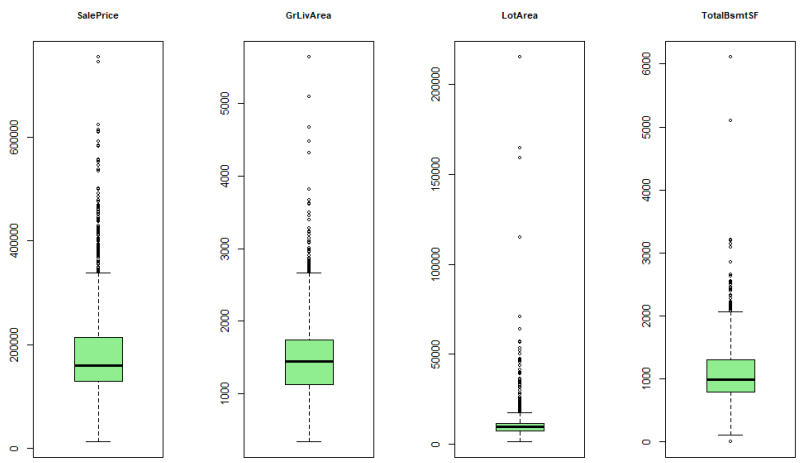

Figure 3 - Boxplots of SalePrice, GrLivArea, LotArea, and TotalBsmtSF Showing Outlier Presence

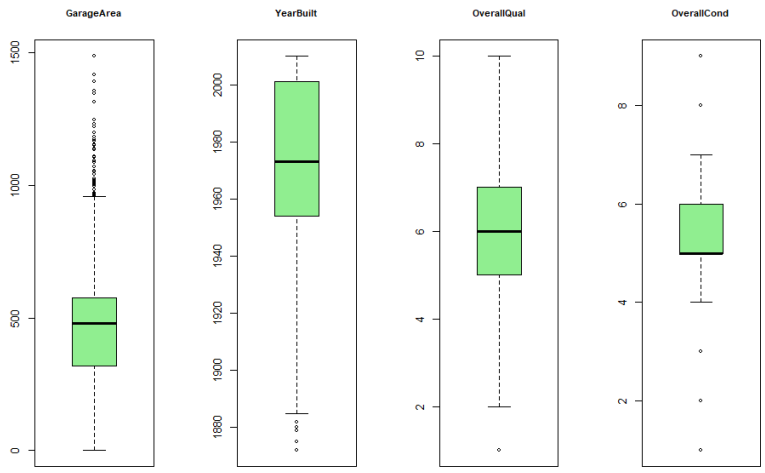

Figure 2 - - Boxplots of GarageArea, YearBuilt, OverallQual, and OverallCond Showing Outlier Presence

The boxplot analysis in Figures 2 and 3 confirms the presence of outliers across multiple variables. SalePrice contains numerous upper outliers representing luxury properties, with values exceeding $400,000 indicated as statistical outliers.

GrLivArea shows several properties with living areas above 4,000 square feet that warrant special attention. LotArea exhibits the most extreme outliers, with some lots exceeding 100,000 square feet compared to a median around 10,000 square feet. These outliers were carefully assessed during model development, with attention to their influence on regression coefficients.

Figure 4 provides a detailed view of the target variable, SalePrice. The histogram confirms the right-skewed distribution with a peak around $130,000-$150,000 and a gradual decrease toward higher values. The boxplot identifies many observations as upper outliers, representing properties that sold for significantly more than typical market values. The median sale price of

Figure 4 - Detailed Distribution Analysis of Target Variable

approximately $160,000 lies below the mean of approximately $180,921, further confirming positive skewness.

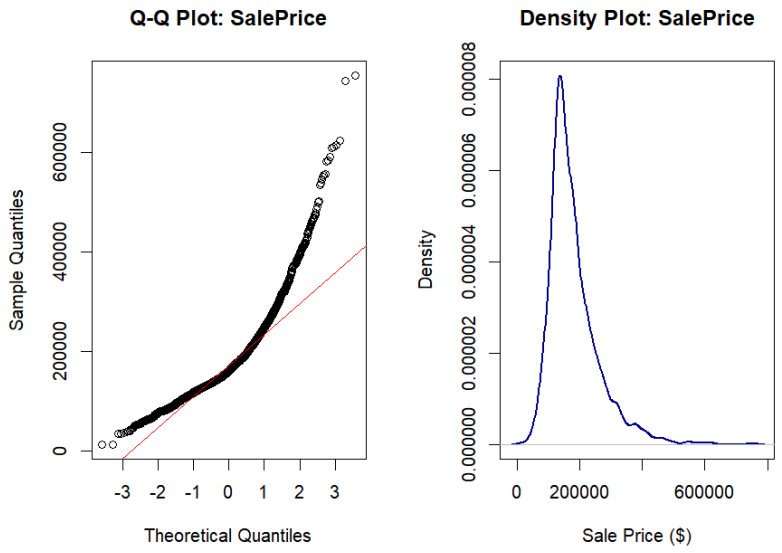

Figure 5 - Q-Q Plot and Density Plot of SalePrice

The Q-Q plot in Figure 5 assesses normality of the SalePrice distribution. The central portion of the data aligns reasonably well with the theoretical normal line, but substantial deviations occur in both tails. The lower tail shows slight deviation, while the upper tail curves sharply upward, indicating heavy right-tail behavior. This pattern suggests that a log transformation might improve normality, though the untransformed values were retained for interpretability. The density plot confirms the unimodal, right-skewed distribution with a smooth peak around $150,000.

Correlation Analysis

The correlation matrix in Figure 6 visualizes relationships among key numeric variables and SalePrice. Circle size and color intensity indicate correlation strength, with larger, darker blue circles representing stronger positive correlations.

SalePrice shows strong positive correlations with OverallQual (r = 0.80), GrLivArea (r = 0.70), GarageArea (r = 0.64), TotalBsmtSF (r = 0.63), and YearBuilt (r = 0.56). Notably, OverallCond shows minimal correlation with SalePrice (r = -0.10), suggesting that condition rating alone does not strongly predict sale price when quality is considered.

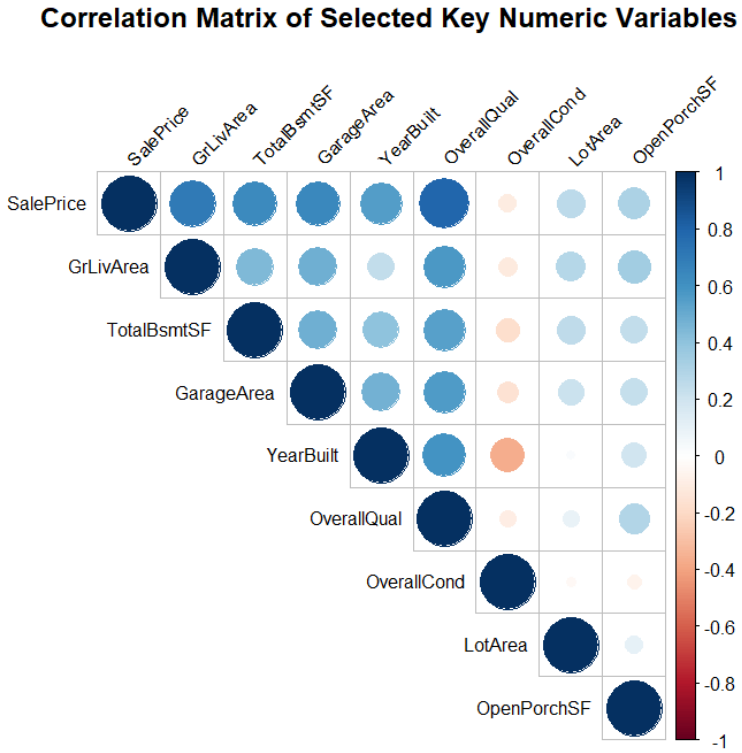

Figure 6 - Correlation Matrix of Selected Key Numeric Variables

Scatterplot Comparison

Figure 7 displays the relationship between SalePrice and OverallQual, which exhibits the highest correlation (r = 0.80) among all predictors. The scatter plot reveals a clear positive linear relationship, as overall quality rating increases from 1 to 10, sale prices increase substantially. At quality rating 5 (average), prices cluster around $100,000-$150,000, while properties rated 8-10 typically sell for $300,000-$500,000 or more. The regression line captures this strong positive trend.

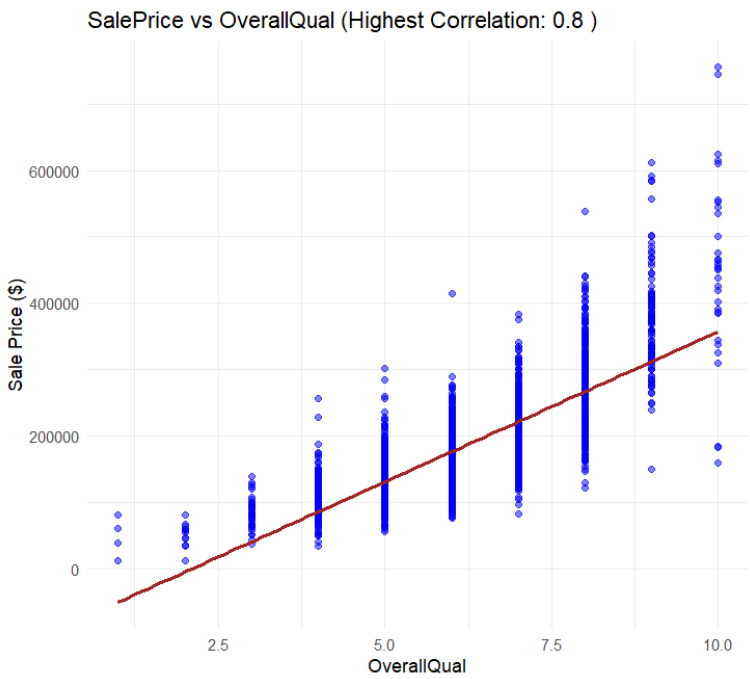

Figure 7 - Scatter Plot of SalePrice vs OverallQual (Highest Correlation: r = 0.80)

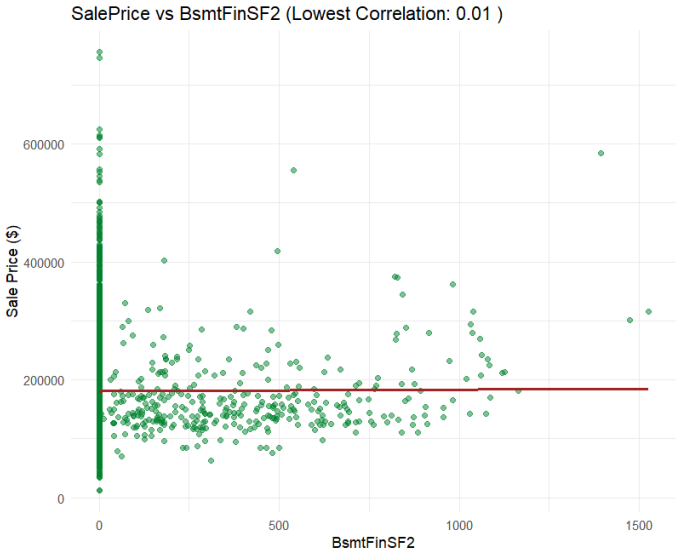

Figure 8 - Scatter Plot of SalePrice vs BsmtFinSF2 (Lowest Correlation: r = 0.01)

Figure 8 illustrates the relationship between SalePrice and BsmtFinSF2 (Type 2 finished basement square footage), which shows the weakest correlation (r = 0.01) with the target variable. The scatter plot displays an essentially horizontal regression line with no evident linear pattern. Points scatter randomly across all sale price ranges regardless of BsmtFinSF2 values, with most observations clustered at zero. This variable provides no predictive value for sale price and was excluded from regression models.

Figure 9 shows the relationship between SalePrice and TotRmsAbvGrd (total rooms above grade), representing a moderate correlation (r = 0.50). The positive linear relationship is evident but with considerable scatter around the trend line. Houses with 5-7 rooms typically sell for $100,000-$200,000, while those with 10+ rooms can exceed $400,000. The discrete nature of room counts creates vertical bands similar to OverallQual. The moderate correlation suggests that room count contributes to price prediction but with less than overall quality or living area.

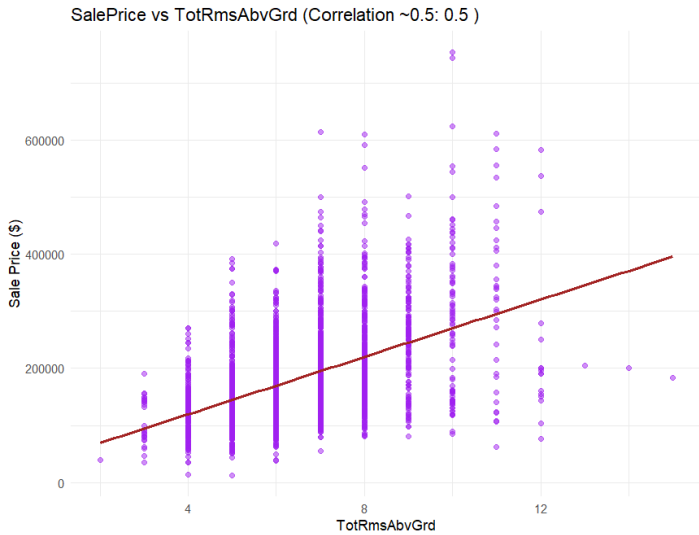

Figure 9 - Scatter Plot of SalePrice vs TotRmsAbvGrd (Correlation: r = 0.50)

The scatter plot comparison clearly demonstrates how correlation strength manifests visually, strong correlations produce tight clustering around the regression line, moderate correlations show a visible trend with significant scatter, and near-zero correlations appear as random clouds with horizontal trend lines.

Initial Regression Model (Model 1)

Based on the correlation analysis and theoretical considerations about housing price determinants, an initial multiple linear regression model was constructed using 4 continuous predictors

- OverallQual (overall quality rating)
- GrLivArea (above-ground living area)
- GarageArea (garage size)
- and HouseAge (calculated as YrSold minus YearBuilt)

The estimated regression equation is expressed as

SalePrice = -62480.58 + 23743.79 * OverallQual + 55.81 * GrLivArea + 60.77 * GarageArea + -380.84 * HouseAge

The model demonstrates strong predictive capability with an R-squared value of 0.7703, indicating that approximately 77.03% of the variance in sale prices is explained by these four predictors. The adjusted R-squared of 0.77 confirms that the model's explanatory power is not artificially inflated by the number of predictors. All coefficients are statistically significant at the $p < 0.001$ level.

The estimated coefficients from the multiple linear regression model provide insight into how each predictor variable is associated with housing sale prices, holding all other variables constant.

Overall Quality (OverallQual).

The strongest correlation with sale price among the predictors. The estimated coefficient indicates that a one-unit increase in the overall quality rating (on a 1–10 scale) is associated with an average increase of approximately $23,744 in sale price. This result is economically meaningful and aligns with expectations, as higher-quality construction and finishes substantially increase property value.

Above-Ground Living Area (GrLivArea).

The coefficient for above-ground living area suggests that for each additional square foot of living space, the sale price increases by approximately $55.81. This positive relationship reflects the market premium placed on larger homes and confirms the importance of living area as a key factor of housing prices.

Garage Area (GarageArea).

The estimated coefficient means that each additional square foot of garage space is associated with an increase of approximately $60.77 in sale price. This result suggests that garage capacity and size contribute meaningfully to perceived property value.

House Age (HouseAge).

House age exhibits a negative relationship with sale price. Specifically, each additional year of age is associated with an average decrease of approximately $380.84 in sale price. This negative coefficient is intuitive, as older homes may require more maintenance and may lack modern design features, reducing their market value relative to newer properties.

Diagnostic Evaluation

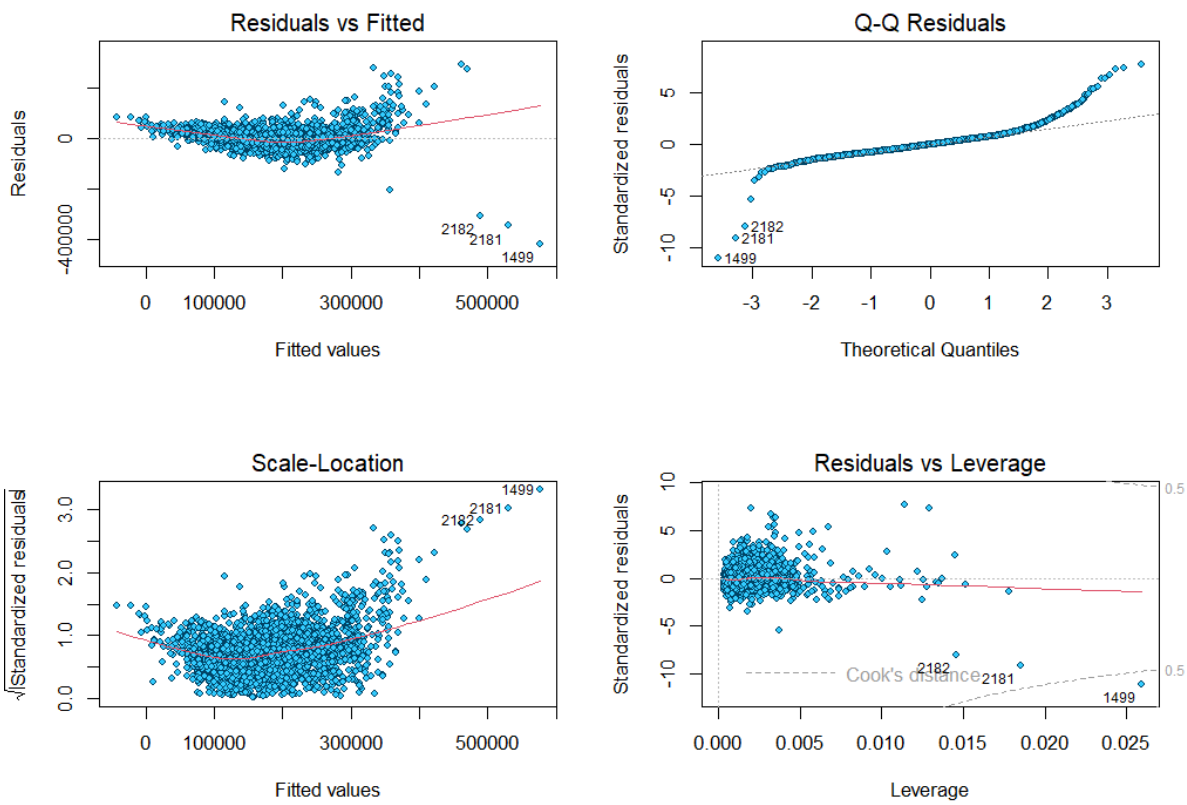

Figure 10 - Diagnostic Plots for Model 1

Residuals vs Fitted

The residuals show a generally random scatter around the horizontal zero line for fitted values up to approximately $300,000, supporting the linearity assumption. However, a slight funnel shape occurs at higher fitted values, with increasing residual spread, suggesting mild heteroscedasticity. Three observations (1499, 2181, 2182) appear as extreme negative residuals

with fitted values around $400,000-$500,000, indicating the model substantially overpredicts their actual sale prices.

Q-Q Plot (Normal Q-Q)

This plot shows normality of residuals. The standardized residuals follow the diagonal line well in the central portion (-2 to +2), indicating approximate normality for the bulk of observations. However, both tails deviate, the left tail curves slightly below the line, and the right tail shows moderate difference above the line. The same three outliers (1499, 2181, 2182) appear in the lower left with standardized residuals below -8, far exceeding typical values.

Scale-Location

This plot provides another view of homoscedasticity by plotting the square root of standardized residuals against fitted values. Preferably, points should scatter randomly with a horizontal red line. The observed pattern shows a slight upward trend from left to right, confirming mild heteroscedasticity where prediction error variance increases with predicted price. This is common in housing data where expensive homes exhibit greater price variability.

Residuals vs Leverage

This plot shows influential observations combining high leverage (unusual predictor values) with large residuals. The dashed curves represent Cook's distance contours at 0.5. Observations 2181 and 2182 appear near the 0.5 contour, indicating substantial influence on model coefficients. Observation 1499 shows the largest negative residual with moderate leverage. These points warrant investigation as potential outliers.

Multicollinearity Assessment

Variance Inflation Factor (VIF) analysis was conducted to assess multicollinearity among predictors. VIF values measure how much the variance of a coefficient estimate is increased due to correlation with other predictors. The conventional levels indicate that VIF values below 5 suggest acceptable collinearity levels, values between 5-10 indicate moderate concern, and values above 10 signal severe multicollinearity requiring remediation.

```
> print(vif_values)
OverallQual   GrLivArea  GarageArea    HouseAge
   2.317481    1.648092    1.670674    1.706947
>
```

Figure 11 - VIF values of model factors

All four predictors demonstrated VIF values well below 5, indicating no multicollinearity concerns. Each predictor provides unique information not captured by the others, supporting their joint inclusion in the model.

If multicollinearity had been detected, potential remediation strategies would include:

1) Removing one of the highly correlated predictors based on theoretical importance,
2) Combining correlated variables through principal component analysis or creating composite indices,
3) Employing ridge regression or LASSO regularization that can handle correlated predictors more effectively by shrinking coefficient estimates, or
4) Centering predictors to reduce structural collinearity in models with interaction terms.

Outlier Analysis

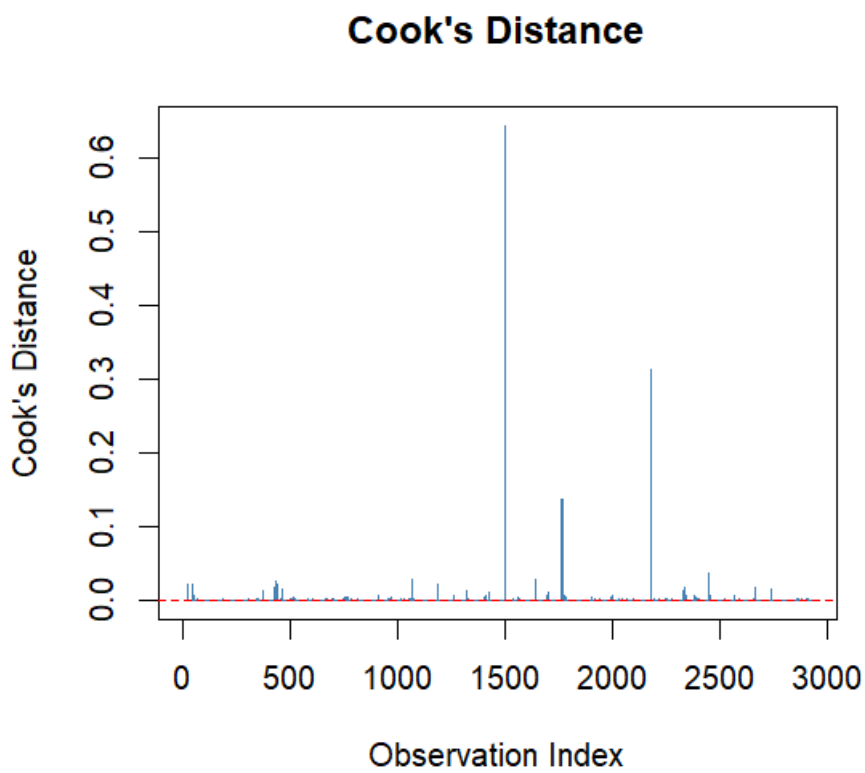

Figure 12 - Cook's Distance Plot Identifying Influential Observations

Figure 12 displays Cook's distance values for all observations. Cook's distance measures the overall influence of each observation on model coefficients by combining leverage and residual magnitude. Three observations stand out dramatically

- Near index 1500 with Cook's distance exceeding 0.6 (observation 1499)
- Near index 1700 with Cook's distance around 0.13
- Near index 2200 with Cook's distance approximately 0.33 (observations 2181-2182).

These values far exceed the 4/n threshold of approximately 0.0014, indicating these observations exert substantial influence on the fitted model.

Multiple diagnostic approaches were employed to identify outliers. Standardized residuals exceeding an absolute value of 3 flagged several observations, with observations 1499, 2181, and 2182 showing standardized residuals below -8, indicating their actual prices fell dramatically below model predictions. Investigation revealed these properties had above-grade living areas exceeding 4,000 square feet but sold for prices much lower than predicted, likely representing partial sales or unusual circumstances as noted in the dataset documentation.

Based on the dataset documentation's recommendation and the diagnostic analysis, properties with above-grade living area exceeding 4,000 square feet were identified for removal. The documentation notes that these represent either true outliers (partial sales not reflecting market values) or unusual sales (very large houses). Removing these observations aligns with the modeling objective of creating a predictive tool for typical residential properties.

Improved Model (Model 2)

Following the diagnostic evaluation, Model 2 was developed by removing extreme outliers identified through the analysis.

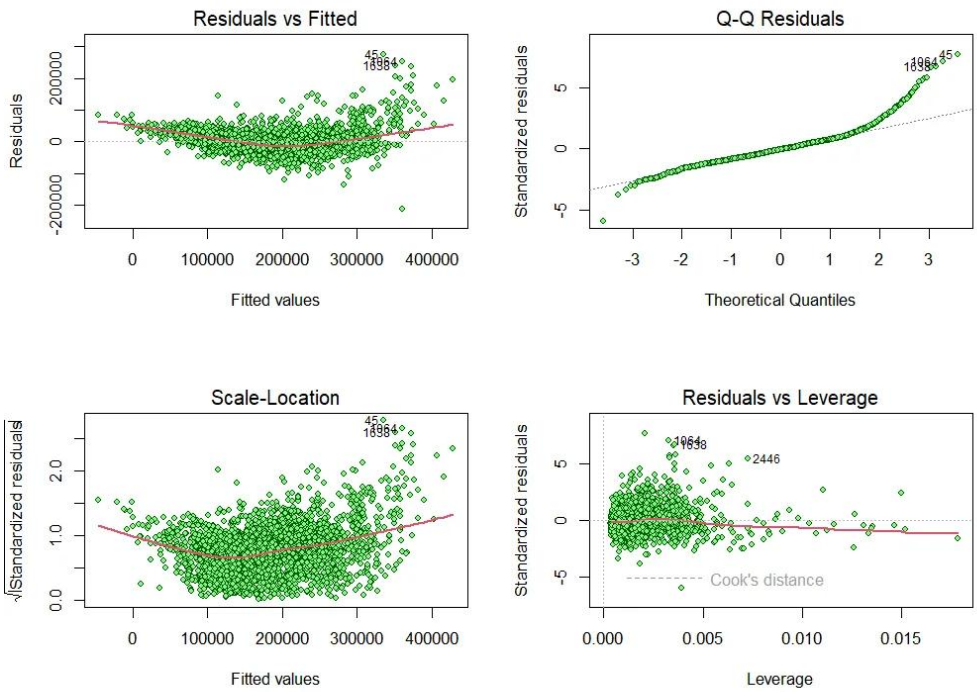

Figure 13 - Diagnostic Plots for Model 2 After Outlier Removal

Figure 13 displays diagnostic plots for Model 2, showing improvement over Model 1. The Residuals vs Fitted plot shows more uniform spread across all fitted values, with reduced extreme outliers. The maximum fitted value is now approximately $400,000 compared to $550,000 in Model 1, reflecting removal of extreme cases. The Q-Q plot shows improved normality, particularly in the lower tail where the extreme negative residuals have been removed. While some deviation remains in the upper tail, the overall alignment with the theoretical line has improved significantly. The Scale-Location plot shows a flatter red line, indicating reduced heteroscedasticity. The Residuals vs Leverage plot no longer shows points near the Cook's distance 0.5 contour, confirming that highly influential observations have been addressed.

	Metric	Model_1	Model_2	Change
1	R-squared	0.7702673	0.7928498	2.93%
2	Adjusted R-squared	0.7699531	0.7925660	2.94%
3	Residual Std Error	38316.2150104	35777.7467588	-6.63%
4	Observations	2930.0000000	2925.0000000	-5

Table 1 - Comparison of Model 1 and Model 2 Performance

Model 2 demonstrates improved performance across all metrics. The adjusted R-squared increased from 0.7699 to 0.7925, indicating slightly better explanatory power. More importantly, the residual standard error decreased from $38,316 to $35,777, a decrease of 6.63%, meaning predictions are more accurate on average. The diagnostic improvements, better residual normality, reduced heteroscedasticity, and elimination of highly influential points provide greater confidence in statistical inference from the model.

All Subsets Regression

To identify the optimal model specification, all subsets regression was performed using the regsubsets function from the leaps package. This search approach systematically evaluates all possible combinations of predictors to identify the best models based on multiple selection criteria.

Ten candidate predictors were considered

- OverallQual
- GrLivArea
- GarageArea
- HouseAge
- TotalBsmtSF
- X1stFlrSF
- FullBath
- TotRmsAbvGrd
- Fireplaces
- LotArea

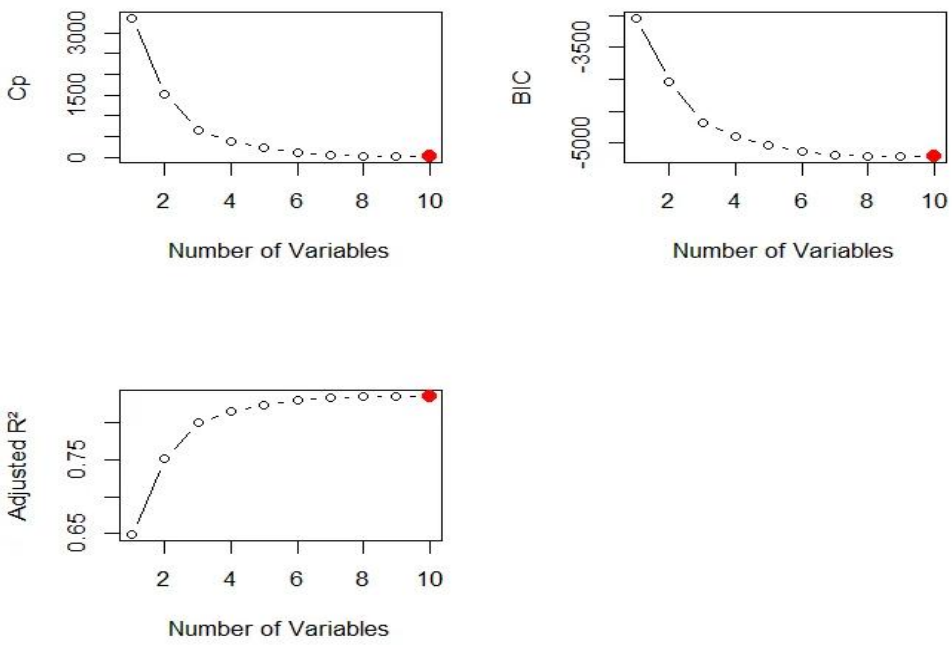

Figure 14 - Cp, BIC, and Adjusted R-squared by Number of Variables

Figure 14 shows three model selection criteria plotted against the number of predictors. The Mallows' Cp plot shows a sharp decline from 1-4 variables, then levels off, with the optimal model (red point) at 10 variables where Cp approaches p (number of parameters).

The BIC plot, which penalizes model complexity more heavily, shows steady decline through 10 variables, selecting the full model as optimal. The Adjusted R-squared plot shows rapid improvement from 1-4 variables, increasing from approximately 0.65 to 0.77, then gradual improvement to approximately 0.79 at 10 variables.

The selection criteria suggest that a model with 7-10 variables provides the best balance. The preferred model based on BIC includes, OverallQual, GrLivArea, GarageArea, HouseAge, TotalBsmtSF, FullBath, Fireplaces, and LotArea.

The model equation is:

$$\boldsymbol{SalePrice = \beta_0 + \beta_1(OverallQual) + \beta_2(GrLivArea) + \beta_3(GarageArea) + \beta_4(HouseAge) + \beta_5(TotalBsmtSF) + \beta_6(FullBath) + \beta_7(Fireplaces) + \beta_8(LotArea)}$$

Model Comparison

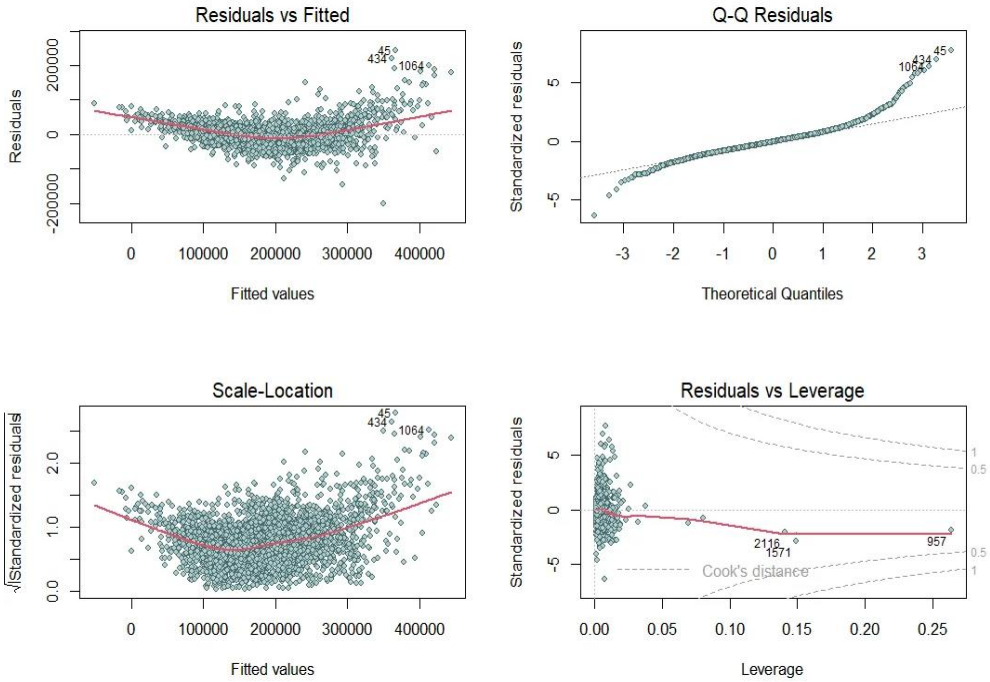

Figure 15 - Diagnostic Plots for Preferred Model from All Subsets Regression

Figure 15 presents diagnostic plots for the preferred model identified through all subsets regression. The Residuals vs Fitted plot shows generally random scatter, though some observations (45, 434, 1064) appear as moderate positive outliers. The Q-Q plot demonstrates good normality in the central region with some deviation in the tails, like previous models. The Scale-Location plot shows relatively constant variance across fitted values. The Residuals vs Leverage plot shows Cook's distance contours at 0.5 and 1.0, with observations 957, 1571, and 2116 appearing near or beyond these thresholds, suggesting some influential points remain that could be investigated further.

	Metric	Model2	Preferred_Model
1	Variables	OverallQual, GrLivArea, GarageArea, HouseAge	OverallQual, GrLivArea, GarageArea, HouseAge, TotalBsmtSF...
2	R-squared	0.7928	0.8369
3	Adj R-squared	0.7926	0.8364
4	AIC	69645.51	68957.86
5	BIC	69681.4	69029.63
6	RSE	35777.75	31777.4

Figure 16 - Model 2 vs Preferred All Subsets Model

The comparison reveals that the preferred model from all subsets regression outperforms Model 2 on most metrics. The adjusted R-squared increases from 0.7926 to 0.8364, a gain of percentage points in explained variance. AIC and BIC both favor the preferred model, indicating better fit even after penalizing for additional complexity. The residual standard error decreases from $35,777 to $31,777, improving prediction accuracy.

However, the choice between models involves trade-offs. Model 2 offers greater parsimony with only 4 predictors, making it easier to interpret and communicate to stakeholders. The preferred model provides slightly better prediction but requires gathering more variables for each prediction.

- For practical applications where interpretability is vital, Model 2 may be preferred.
- For applications prioritizing predictive accuracy, all subsets model is recommended

Conclusion

This study demonstrated a complete and rigorous regression modeling workflow using the Ames Housing dataset, emphasizing both predictive performance and diagnostic validity. Through exploratory data analysis and correlation assessment, key predictors of housing sale prices were identified, including overall quality, above-grade living area, garage area, and house age. An initial regression model using these variables explained approximately 77% of the variation in sale prices, with all coefficients exhibiting expected signs and strong statistical significance.

Comprehensive diagnostic evaluation revealed mild heteroscedasticity and the presence of highly influential observations, particularly properties with above-grade living areas exceeding 4,000 square feet. Guided by residual analysis, Cook's distance, and dataset documentation, these extreme observations were removed to produce an improved model. The refined model demonstrated better diagnostic behavior, reduced residual variance, and improved predictive accuracy, increasing the adjusted R-squared to approximately 0.79 while maintaining interpretability and adherence to regression assumptions.

An automated all subsets regression approach was subsequently applied to explore alternative model specifications. The preferred model identified through this process achieved higher explanatory power and lower information criteria by incorporating additional predictors such as basement area, bathroom count, fireplaces, and lot size. While this model provided superior predictive accuracy, it introduced greater complexity. Ultimately, the analysis highlights the trade-off between model interpretability and predictive performance and underscores the importance of diagnostic evaluation, informed outlier handling, and thoughtful model selection in applied regression analysis.

REFERENCES

De Cock, D. (2011). Ames, Iowa: Alternative to the Boston housing data as an end of semester regression project. Journal of Statistics Education, 19(3), 1-15. https://doi.org/10.1080/10691898.2011.11889627

Fox, J., & Weisberg, S. (2019). An R companion to applied regression (3rd ed.). Sage Publications.

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An introduction to statistical learning with applications in R (2nd ed.). Springer.

Kuiper, S. (2008). Introduction to multiple regression: How much is your car worth? Journal of Statistics Education, 16(3), 1-11.

Lumley, T. (2020). leaps: Regression subset selection (R package version 3.1). https://CRAN.R-project.org/package=leaps

Pardoe, I. (2008). Modeling home prices using realtor data. Journal of Statistics Education, 16(2), 1-9.

R Core Team. (2024). R: A language and environment for statistical computing. R Foundation for Statistical Computing. https://www.R-project.org/

Wei, T., & Simko, V. (2021). corrplot: Visualization of a correlation matrix (R package version 0.92). https://CRAN.R-project.org/package=corrplot

Wickham, H., & Grolemund, G. (2017). R for data science: Import, tidy, transform, visualize, and model data. O'Reilly Media.

APPENDIX A : R CODE

```
## ALY6015 - Regression Diagnostics with R

## Module 1 - Assignment 1



# Reset environment

rm(list = ls())     # clears all objects

cat("\014")         # clears the console



# Install and load necessary packages

#install.packages("tidyverse")

#install.packages("car")

#install.packages("corrplot")

#install.packages("ggcorrplot")

#install.packages("leaps")



library(tidyverse)

library(car)

library(corrplot)

library(ggcorrplot)

library(leaps)



# ------------------------------- 1. DATASET LOADING --------------------------------------------



ames_data <- read.csv("AmesHousing.csv")
```

```
# View the structure of the dataset
glimpse(ames_data)
head(ames_data)
names(ames_data)

# Clean column names by removing dots
names(ames_data) <- gsub("\\.", "", names(ames_data))

ames_data <- ames_data %>%
    select(-Order)





# ---------------------------- 2. EDA ANALYSIS ------------------------------------------------

# Identify variable types
numeric_vars <- ames_data %>% select_if(is.numeric)
categorical_vars <- ames_data %>% select_if(is.character)

# MISSING VALUES ANALYSIS
missing_summary <- data.frame(
    Variable = names(ames_data),
    Missing_Count = colSums(is.na(ames_data)),
    Missing_Percent = round(colSums(is.na(ames_data)) / nrow(ames_data) * 100, 2)
```

```
)
missing_summary <- missing_summary[missing_summary$Missing_Count > 0, ]
missing_summary <- missing_summary[order(-missing_summary$Missing_Count), ]




### NUMERIC VARIABLES ANALYSIS
# Summary statistics for ALL numeric variables
numeric_summary <- data.frame(
  Variable = names(numeric_vars),
  Mean = sapply(numeric_vars, function(x) round(mean(x, na.rm = TRUE), 2)),
  Median = sapply(numeric_vars, function(x) round(median(x, na.rm = TRUE), 2)),
  SD = sapply(numeric_vars, function(x) round(sd(x, na.rm = TRUE), 2)),
  Min = sapply(numeric_vars, function(x) round(min(x, na.rm = TRUE), 2)),
  Max = sapply(numeric_vars, function(x) round(max(x, na.rm = TRUE), 2)),
  Q1 = sapply(numeric_vars, function(x) round(quantile(x, 0.25, na.rm = TRUE), 2)),
  Q3 = sapply(numeric_vars, function(x) round(quantile(x, 0.75, na.rm = TRUE), 2))
)

# Create multi-panel plot for key continuous variables
key_vars <- c("SalePrice", "GrLivArea", "LotArea", "TotalBsmtSF",
              "GarageArea", "YearBuilt", "OverallQual", "OverallCond")
key_vars <- key_vars[key_vars %in% names(ames_data)]

par(mfrow = c(3, 3))
```

```
for (var in key_vars) {
  if (var %in% names(ames_data)) {
    hist(ames_data[[var]],
         main = paste("Distribution of", var),
         xlab = var,
         col = "steelblue",
         breaks = 30,
         cex.main = 0.9)
  }
}
par(mfrow = c(1, 1))

# Boxplots for key continuous variables to identify outliers
par(mfrow = c(1, 4))
for (var in key_vars) {
  if (var %in% names(ames_data)) {
    boxplot(ames_data[[var]],
            main = var,
            col = "lightgreen",
            cex.main = 0.8)
  }
}
par(mfrow = c(1, 1))
```

```
### TARGET VARIABLE (SALEPRICE) DETAILED ANALYSIS
saleprice_stats_wide <- ames_data %>%
  summarise(
    Mean = round(mean(SalePrice, na.rm = TRUE), 2),
    Median = round(median(SalePrice, na.rm = TRUE), 2),
    Std_Dev = round(sd(SalePrice, na.rm = TRUE), 2),
    Min = round(min(SalePrice, na.rm = TRUE), 2),
    Max = round(max(SalePrice, na.rm = TRUE), 2),
    Range = round(Max - Min, 2),
    CV_Percent = round(sd(SalePrice, na.rm = TRUE) / mean(SalePrice, na.rm = TRUE) *
100, 2),
    Skewness = if (require(moments, quietly = TRUE)) round(skewness(SalePrice, na.rm =
TRUE), 2) else NA,
    Kurtosis = if (require(moments, quietly = TRUE)) round(kurtosis(SalePrice, na.rm =
TRUE), 2) else NA,
    Q1 = round(quantile(SalePrice, 0.25, na.rm = TRUE), 2),
    Q3 = round(quantile(SalePrice, 0.75, na.rm = TRUE), 2),
    IQR = round(Q3 - Q1, 2)
  )

# Prevent scientific notation
options(scipen = 999)

# SalePrice visualizations
par(mfrow = c(1, 2))
```

```
# Histogram
hist(ames_data$SalePrice,
     main = "Distribution of Sale Prices",
     xlab = "Sale Price ($)",
     col = "lightblue",
     breaks = 50)

# Boxplot
boxplot(ames_data$SalePrice,
        main = "Boxplot of Sale Prices",
        ylab = "Sale Price ($)",
        col = "lightgreen")

# Q-Q plot for normality
qqnorm(ames_data$SalePrice, main = "Q-Q Plot: SalePrice")
qqline(ames_data$SalePrice, col = "red")

# Density plot
plot(density(ames_data$SalePrice, na.rm = TRUE),
     main = "Density Plot: SalePrice",
     xlab = "Sale Price ($)",
     col = "darkblue",
     lwd = 2)

par(mfrow = c(1, 1))
```

```
### CATEGORICAL VARIABLES ANALYSIS

# Analyze key categorical variables
key_categorical <- c("MSZoning", "Neighborhood", "BldgType", "HouseStyle",
                     "OverallQual", "OverallCond", "HeatingQC", "CentralAir")
key_categorical <- key_categorical[key_categorical %in% names(ames_data)]

for (var in key_categorical) {
  if (var %in% names(ames_data)) {
    cat("---", var, "---\n")
    freq_table <- table(ames_data[[var]], useNA = "ifany")
    freq_df <- data.frame(
      Category = names(freq_table),
      Count = as.numeric(freq_table),
      Percentage = round(as.numeric(freq_table) / sum(freq_table) * 100, 2)
    )
    freq_df <- freq_df[order(-freq_df$Count), ]
    print(freq_df, row.names = FALSE)
    cat("\n")
  }
}

# Visualize key categorical variables
```

```
# Select top categorical variables for visualization
viz_categorical <- c("BldgType", "HouseStyle","OverallQual", "OverallCond", "HeatingQC",
"CentralAir")
viz_categorical <- viz_categorical[viz_categorical %in% names(ames_data)]


my_colors <- c("#4E79A7", "#F28E2B", "#E15759", "#76B7B2", "#59A14F", "#EDC948",
"#B07AA1", "#FF9DA7", "#9C755F", "#BAB0AC")


par(mfrow = c(2, 3))
for (var in viz_categorical) {
  if (var %in% names(ames_data)) {
    freq <- table(ames_data[[var]])
    barplot(freq,
            main = paste("Distribution of", var),
            col = my_colors[1:length(freq)],
            las = 2,
            cex.names = 0.7,
            cex.main = 0.9)
  }
}
par(mfrow = c(1, 1))



### RELATIONSHIP BETWEEN CATEGORICAL AND TARGET VARIABLE
# Boxplots showing SalePrice by categorical variables
```

```
analysis_categorical <- c("BldgType", "HouseStyle","OverallQual", "OverallCond", "HeatingQC", "CentralAir")

analysis_categorical <- analysis_categorical[analysis_categorical %in% names(ames_data)]


par(mfrow = c(1, 2))

for (var in analysis_categorical) {

  if (var %in% names(ames_data)) {

    boxplot(SalePrice ~ ames_data[[var]],

            data = ames_data,

            main = paste("SalePrice by", var),

            xlab = var,

            ylab = "Sale Price ($)",

            col = "lightcoral",

            las = 2,

            cex.axis = 0.6,

            cex.main = 0.9)

  }

}

par(mfrow = c(1, 1))



# YEAR/TEMPORAL ANALYSIS

# Year Built distribution

if ("YearBuilt" %in% names(ames_data)) {

  par(mfrow = c(1, 2))
```

```
  hist(ames_data$YearBuilt,
      main = "Distribution of Year Built",
      xlab = "Year",
      col = "skyblue",
      breaks = 30)

  plot(ames_data$YearBuilt, ames_data$SalePrice,
      main = "SalePrice vs Year Built",
      xlab = "Year Built",
      ylab = "Sale Price ($)",
      col = alpha("blue", 0.3),
      pch = 19)
  par(mfrow = c(1, 1))
}

# Sale Year and Month analysis
if ("YrSold" %in% names(ames_data) && "MoSold" %in% names(ames_data)) {
  print(table(ames_data$YrSold))
  print(table(ames_data$MoSold))

  par(mfrow = c(1, 2))
  barplot(table(ames_data$YrSold),
          main = "Sales by Year",
          xlab = "Year",
          ylab = "Count",
```

```
        col = "coral")

  barplot(table(ames_data$MoSold),
        main = "Sales by Month",
        xlab = "Month",
        ylab = "Count",
        col = "lightgreen")
  par(mfrow = c(1, 1))
}





# ------------------------------ 3. IMPUTE MISSING VALUES -----------------------------------

# Check for missing value percentage
missing_summary <- data.frame(
  Variable = names(ames_data),
  Missing_Count = colSums(is.na(ames_data)),
  Missing_Percent = round(colSums(is.na(ames_data)) / nrow(ames_data) * 100, 2)
)
missing_summary <- missing_summary[missing_summary$Missing_Count > 0, ]

# Create a copy of the dataset for cleaning
ames_clean <- ames_data
```

```
# Create Age variable instead of using YearBuilt
if ("YearBuilt" %in% names(ames_clean) && "YrSold" %in% names(ames_clean)) {
  ames_clean$HouseAge <- ames_clean$YrSold - ames_clean$YearBuilt
}

# Remove Variables with more than 30% missing values
ames_clean <- ames_clean %>%
  select(-Alley, -FireplaceQu, -PoolQC, -Fence, -MiscFeature)

# Impute missing values for numeric variables with their mean
numeric_columns <- names(ames_clean)[sapply(ames_clean, is.numeric)]

for (col in numeric_columns) {
  if (sum(is.na(ames_clean[[col]])) > 0) {
    mean_val <- mean(ames_clean[[col]], na.rm = TRUE)
    ames_clean[[col]][is.na(ames_clean[[col]])] <- mean_val
    print(col)
  }
}
colSums(is.na(ames_clean))
numeric_vars <- ames_clean %>% select_if(is.numeric)
```

```
# ------------------------------- 4. CORRELATION MATRIX -------------------------------------

# Remove identifiers
numeric_vars <- numeric_vars %>% select(-PID)

# Calculate correlation matrix
cor_matrix <- cor(numeric_vars, use = "complete.obs")

# Display correlations with SalePrice (sorted)
saleprice_cor <- cor_matrix[, "SalePrice"]
saleprice_cor_sorted <- sort(abs(saleprice_cor), decreasing = TRUE)

print(saleprice_cor_sorted)



# ---------------------------------- 5. CORRELATION MATRIX VISUALIZATION ---------------
-----------
# Correlation plot
corrplot(
  cor_matrix,
  method = "circle",
  type = "upper",
```

```
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 0.5,
  title = "Correlation Matrix of Key Numeric Variables",
  mar = c(0, 0, 1, 0)
)

# Focused correlation plot for key variables
key_numeric_vars <- numeric_vars %>%
  select(SalePrice, GrLivArea, TotalBsmtSF, GarageArea, YearBuilt, OverallQual, OverallCond, LotArea, OpenPorchSF)

cor_matrix_key <- cor(key_numeric_vars, use = "complete.obs")
corrplot(
  cor_matrix_key,
  method = "circle",
  type = "upper",
  tl.col = "black",
  tl.srt = 45,
  tl.cex = 0.8,
  title = "Correlation Matrix of Selected Key Numeric Variables",
  mar = c(0, 0, 1, 0)
)
```

```
# ---------------------------------- 6. SCATTER PLOTS FOR KEY RELATIONSHIPS ----------
----------
# Identify variables with different correlation strengths with SalePrice
saleprice_cor_df <- data.frame(
    Variable = names(saleprice_cor),
    Correlation = saleprice_cor
)
saleprice_cor_df <- saleprice_cor_df[saleprice_cor_df$Variable != "SalePrice", ]
saleprice_cor_df <- saleprice_cor_df[order(abs(saleprice_cor_df$Correlation), decreasing =
TRUE), ]

# Highest correlation
highest_cor_var <- saleprice_cor_df$Variable[1]
highest_cor_val <- saleprice_cor_df$Correlation[1]

# Lowest correlation (closest to zero, but not zero)
lowest_cor_var <- saleprice_cor_df$Variable[nrow(saleprice_cor_df)]
lowest_cor_val <- saleprice_cor_df$Correlation[nrow(saleprice_cor_df)]

# Correlation closest to 0.5
closest_to_05 <- saleprice_cor_df[which.min(abs(abs(saleprice_cor_df$Correlation) - 0.5)), ]
mid_cor_var <- closest_to_05$Variable
mid_cor_val <- closest_to_05$Correlation
```

```
# Scatter plots for these variables
ggplot(numeric_vars, aes(x = .data[[highest_cor_var]], y = SalePrice)) +
  geom_point(color = "blue", alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, color = "#A32424") +
  labs(title = paste("SalePrice vs", highest_cor_var, "(Highest Correlation:", round(highest_cor_val, 2), ")"),
       x = highest_cor_var, y = "Sale Price ($)") +
  theme_minimal()

ggplot(numeric_vars, aes(x = .data[[lowest_cor_var]], y = SalePrice)) +
  geom_point(color = "#00832D", alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, color = "#A32424") +
  labs(title = paste("SalePrice vs", lowest_cor_var, "(Lowest Correlation:", round(lowest_cor_val, 2), ")"),
       x = lowest_cor_var, y = "Sale Price ($)") +
  theme_minimal()

ggplot(numeric_vars, aes(x = .data[[mid_cor_var]], y = SalePrice)) +
  geom_point(color = "purple", alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE, color = "#A32424") +
  labs(title = paste("SalePrice vs", mid_cor_var, "(Correlation ~0.5:", round(mid_cor_val, 2), ")"),
       x = mid_cor_var, y = "Sale Price ($)") +
  theme_minimal()
```

```
# ------------------------------- 7. FIT A REGRESSION MODEL ------------------------------------
-------


# Continuous variables with good correlations Based on correlation analysis

# 1. OverallQual (overall quality rating)

# 2. GrLivArea  (above grade living area)

# 3. GarageArea (garage area)

# 4. HouseAge (age of house - calculated from YrSold - YearBuilt)



# Model 1 : Fit the multiple linear regression model

model1 <- lm(SalePrice ~ OverallQual + GrLivArea + GarageArea + HouseAge,

             data = ames_clean)


# Display model summary

summary(model1)


# Extract coefficients

coef_model1 <- coef(model1)

print(coef_model1)


cat("SalePrice =", round(coef_model1[1], 2), "+",
```

```
  round(coef_model1[2], 2), "* OverallQual +",
  round(coef_model1[3], 2), "* GrLivArea +",
  round(coef_model1[4], 2), "* GarageArea +",
  round(coef_model1[5], 2), "* HouseAge\n")


# ------------ 8.Report Model in Equation Form and Interpret Coefficients -----------------

cat("\nCoefficient Interpretations:\n")

cat("- OverallQual:", round(coef_model1[2], 2), "\n")
cat("  For each 1-unit increase in overall quality rating (on a 1-10 scale), \n")
cat("   sale price increases by $", round(coef_model1[2], 2), ", holding other variables constant\n\n")

cat("- GrLivArea:", round(coef_model1[3], 2), "\n")
cat("  For each additional square foot of living area, sale price increases by $",
  round(coef_model1[3], 2), ", holding other variables constant\n\n")

cat("- GarageArea:", round(coef_model1[4], 2), "\n")
cat("  For each additional square foot of garage area, sale price increases by $",
  round(coef_model1[4], 2), ", holding other variables constant\n\n")

cat("- HouseAge:", round(coef_model1[5], 2), "\n")
if (coef_model1[5] < 0) {
```

```
  cat("  For each additional year of age, sale price DECREASES by $",
      abs(round(coef_model1[5], 2)), ", holding other variables constant\n")
  cat("  This negative coefficient makes sense: older houses typically sell for less\n")
} else {
  cat("  For each additional year of age, sale price increases by $",
      round(coef_model1[5], 2), ", holding other variables constant\n")
  cat("  Note: This unexpected positive coefficient may indicate confounding or that\n")
  cat("  older houses in this dataset have been renovated/maintained exceptionally well\n")
}


# -------------------------------- 9. PLOT REGRESSION MODEL --------------------------------
# Standard regression diagnostics
par(mfrow = c(2, 2))
# Residuals vs Fitted plot
plot(model1,
    which = 1,
    pch = 21,
    bg = "#33ccff",
    col = "#004466",
    cex = 0.8)


# Q-Q plot
plot(model1,
    which = 2,
    pch = 21,
```

```
    bg = "#33ccff",
    col = "#004466",
    cex = 0.8)

# Scale-Location plot
plot(model1,
    which = 3,
    pch = 21,
    bg = "#33ccff",
    col = "#004466",
    cex = 0.8)

# Residuals vs Leverage plot
plot(model1,
    which = 5,
    pch = 21,
    bg = "#33ccff",
    col = "#004466",
    cex = 0.8)
par(mfrow = c(1, 1))

# Extract fitted values and residuals
ames_clean$Predicted_SalePrice <- fitted(model1)
ames_clean$Residuals <- residuals(model1)
# Observed vs Predicted plot
```

```
ggplot(ames_clean, aes(x = Predicted_SalePrice, y = SalePrice)) +
  geom_point(
    shape = 21,
    fill = "#70dbdb",
    color = "#0a2929",
    alpha = 0.5
  ) +
  geom_abline(slope = 1, intercept = 0, color = "#002699", linetype = "dashed",linewidth=0.8)
+
  labs(title = "Observed vs Predicted SalePrice",
     x = "Predicted SalePrice ($)",
     y = "Observed SalePrice ($)") +
  theme_minimal()


# ---------------------------- 10. CHECK FOR MULTICOLLINEARITY Check -------------------
------
# Calculate VIF
vif_values <- vif(model1)
print(vif_values)


# Check if multicollinearity exists
if (any(vif_values > 10)) {
  cat("SEVERE MULTICOLLINEARITY DETECTED!\n")
  cat("Variables with VIF > 10:", names(vif_values[vif_values > 10]), "\n")
} else if (any(vif_values > 5)) {
```

```
    cat("MODERATE MULTICOLLINEARITY DETECTED\n")
    cat("Variables with VIF > 5:", names(vif_values[vif_values > 5]), "\n")
} else {
    cat("No significant multicollinearity detected. All VIF values < 5.\n")
}




# ---------------------------- 11. CHECK FOR OUTLIERS ----------------------------------------


# Standardized residuals
std_residuals <- rstandard(model1)


# Identify outliers
outliers_index <- which(abs(std_residuals) > 3)


if (length(outliers_index) > 0) {
    cat("Outliers based on standardized residuals (|z| > 3):\n")
    print(ames_clean[outliers_index, c("SalePrice", "OverallQual", "GrLivArea",
"GarageArea", "YearBuilt")])
}


# Cook's Distance: influential points
cooks_d <- cooks.distance(model1)
influential <- which(cooks_d > 4 / nrow(ames_clean))
```

```
cat("\nNumber of influential points (Cook's D > 4/n):", length(influential), "\n")
if (length(influential) > 0) print(head(influential, 10))

# Leverage: high leverage points
lev <- hatvalues(model1)
high_lev <- which(lev > 2 * mean(lev))

cat("\nNumber of high leverage points (h > 2*mean(h)): ", length(high_lev), "\n")

# Combined quick plot of Cook's distance
plot(cooks_d, type = "h", col = "steelblue",
     main = "Cook's Distance",
     xlab = "Observation Index",
     ylab = "Cook's Distance")
abline(h = 4 / nrow(ames_clean), col = "red", lty = 2)

# ------------------ 12.CORRECT ISSUES AND IMPROVE MODEL  ------------------------------
---

# Remove extreme outliers
# Based on the documentation, remove houses > 4000 sq ft
ames_clean2 <- ames_clean[ames_clean$GrLivArea <= 4000, ]
cat("Removed", nrow(ames_clean) - nrow(ames_clean2), "observations with GrLivArea >
4000 sq ft\n")
```

```
# Remove observations with high Cook's distance
if (length(influential) > 0 && length(influential) < 100) {
  ames_clean2 <- ames_clean2[-influential[influential %in% rownames(ames_clean2)], ]
  cat("Removed", length(influential), "influential observations\n")
}

# Model 2: Refit the model
model2 <- lm(SalePrice ~ OverallQual + GrLivArea + GarageArea + HouseAge,
             data = ames_clean2)

#Model 2 Summary (After Corrections)
summary(model2)

# Compare models
model_comparison <- data.frame(
  Metric = c(
    "R-squared",
    "Adjusted R-squared",
    "Residual Std Error",
    "Observations"
  ),
  Model_1 = c(
    summary(model1)$r.squared,
    summary(model1)$adj.r.squared,
```

```
      summary(model1)$sigma,
      nobs(model1)
   ),
   Model_2 = c(
      summary(model2)$r.squared,
      summary(model2)$adj.r.squared,
      summary(model2)$sigma,
      nobs(model2)
   )
)

model_comparison$Change <- c(
   paste0(
      round(
         (model_comparison$Model_2[1:3] - model_comparison$Model_1[1:3]) /
            model_comparison$Model_1[1:3] * 100,
         2
      ),
      "%"
   ),
   model_comparison$Model_2[4] - model_comparison$Model_1[4]
)


print(model_comparison)
```

```
if (model_comparison$Adjusted_R_Squared[2] >
    model_comparison$Adjusted_R_Squared[1]) {

    cat("Model 2 shows an improvement, with a higher adjusted R-squared than Model 1.\n")

} else {

    cat("Model 2 does not improve upon Model 1; adjusted R-squared is similar or lower.\n")
}



# Diagnostic plots for improved model2
par(mfrow = c(2, 2))
# Residuals vs Fitted plot
plot(model2,
     which = 1,
     pch = 21,
     bg = "lightgreen",
     col = "darkgreen",
     cex = 0.8,
     lwd = 2)

# Q-Q plot
plot(model2,
```

```
    which = 2,
    pch = 21,
    bg = "lightgreen",
    col = "darkgreen",
    cex = 0.8)

# Scale-Location plot
plot(model2,
    which = 3,
    pch = 21,
    bg = "lightgreen",
    col = "darkgreen",
    cex = 0.8,
    lwd = 2)

# Residuals vs Leverage plot
plot(model2,
    which = 5,
    pch = 21,
    bg = "lightgreen",
    col = "darkgreen",
    cex = 0.8,
    lwd = 2)

par(mfrow = c(1, 1))
```

```
# ---------------------------- 13. SUBSETS REGRESSION ------------------------------------------
--

# Prepare data for all subsets regression
# Select potential predictors
predictors <- c("OverallQual", "GrLivArea", "GarageArea", "HouseAge",
                "TotalBsmtSF", "X1stFlrSF", "FullBath", "TotRmsAbvGrd",
                "Fireplaces", "LotArea")

# Ensure all predictors exist in dataset
predictors <- predictors[predictors %in% names(ames_clean2)]
print(predictors)

# Create formula
formula_str <- paste("SalePrice ~", paste(predictors, collapse = " + "))

# Run all subsets regression
regfit_full <- regsubsets(as.formula(formula_str),
                          data = ames_clean2,
                          nvmax = length(predictors))
```

```
# Summary
reg_summary <- summary(regfit_full)

# Find best model based on different criteria
best_cp <- which.min(reg_summary$cp)
best_bic <- which.min(reg_summary$bic)
best_adjr2 <- which.max(reg_summary$adjr2)

cat("\nBest model by Cp:", best_cp, "variables\n")
cat("Best model by BIC:", best_bic, "variables\n")
cat("Best model by Adjusted R²:", best_adjr2, "variables\n")

# Plot selection criteria
par(mfrow = c(2, 2))
plot(reg_summary$cp, xlab = "Number of Variables", ylab = "Cp", type = "b")
points(best_cp, reg_summary$cp[best_cp], col = "red", cex = 2, pch = 20)

plot(reg_summary$bic, xlab = "Number of Variables", ylab = "BIC", type = "b")
points(best_bic, reg_summary$bic[best_bic], col = "red", cex = 2, pch = 20)

plot(reg_summary$adjr2, xlab = "Number of Variables", ylab = "Adjusted R²", type = "b")
points(best_adjr2, reg_summary$adjr2[best_adjr2], col = "red", cex = 2, pch = 20)

par(mfrow = c(1, 1))
```

```
# Get coefficients of best model (using BIC as criterion)

best_model_vars <- names(coef(regfit_full, best_bic))[-1]

cat("\nBest Model Variables (by BIC):\n")

print(best_model_vars)


# Model 3 : Fit the preferred model

formula_best <- paste("SalePrice ~", paste(best_model_vars, collapse = " + "))

model_preferred <- lm(as.formula(formula_best), data = ames_clean2)


cat("\nPreferred Model Summary:\n")

summary(model_preferred)


# Report equation

coef_preferred <- coef(model_preferred)

cat("\n=== PREFERRED MODEL EQUATION ===\n")

equation_parts <- paste(names(coef_preferred)[-1], "*", round(coef_preferred[-1], 2))

cat("SalePrice =", round(coef_preferred[1], 2), "+", paste(equation_parts, collapse = " + "),
"\n")




# ---------------------------- 14. COMPARE THE MODELS ----------------------------------------
---


# Model Comparison
```

```
cat("\nModel 2 (Improved Model 1):\n","Variables:", paste(names(coef(model2))[-1], collapse = ", "), "\n","R-squared:", summary(model2)$r.squared, "\n","Adj R-squared:", summary(model2)$adj.r.squared, "\n","AIC:", AIC(model2), "\n","BIC:", BIC(model2), "\n","RSE:", summary(model2)$sigma, "\n\n")

cat("Model 3 Preferred Model (All Subsets):\n","Variables:", paste(names(coef(model_preferred))[-1], collapse = ", "), "\n","R-squared:", summary(model_preferred)$r.squared, "\n","Adj R-squared:", summary(model_preferred)$adj.r.squared, "\n","AIC:", AIC(model_preferred), "\n","BIC:", BIC(model_preferred), "\n","RSE:", summary(model_preferred)$sigma, "\n\n")


# Detailed Model Comparison Table

final_comparison <- data.frame(

  Metric = c("Variables", "R-squared", "Adj R-squared", "AIC", "BIC", "RSE"),

  Model2 = c(

    paste(names(coef(model2))[-1], collapse = ", "),

    round(summary(model2)$r.squared, 4),

    round(summary(model2)$adj.r.squared, 4),

    round(AIC(model2), 2),

    round(BIC(model2), 2),

    round(summary(model2)$sigma, 2)

  ),

  Preferred_Model = c(

    paste(names(coef(model_preferred))[-1], collapse = ", "),

    round(summary(model_preferred)$r.squared, 4),

    round(summary(model_preferred)$adj.r.squared, 4),

    round(AIC(model_preferred), 2),

    round(BIC(model_preferred), 2),
```

```
        round(summary(model_preferred)$sigma, 2)
    )
)
print(final_comparison)


# Model Preference Justification
if (summary(model_preferred)$adj.r.squared > summary(model2)$adj.r.squared) {
    cat("PREFERRED: All Subsets Model\n","Reason: Higher Adjusted R-squared and better model selection criteria\n")
} else {
    cat("PREFERRED: Model 2\n","Reason: Similar performance with fewer variables (parsimony principle)\n")
}




# ==================================================================================
# FINAL DIAGNOSTIC PLOTS FOR PREFERRED MODEL
# ==================================================================================


par(mfrow = c(2, 2))
# Residuals vs Fitted plot
plot(model_preferred,
```

```
    which = 1,
    pch = 21,
    bg = "#b3cccc",
    col = "#476b6b",
    cex = 0.8,
    lwd = 2)

# Q-Q plot
plot(model_preferred,
    which = 2,
    pch = 21,
    bg = "#b3cccc",
    col = "#476b6b",
    cex = 0.8)

# Scale-Location plot
plot(model_preferred,
    which = 3,
    pch = 21,
    bg = "#b3cccc",
    col = "#476b6b",
    cex = 0.8,
    lwd = 2)

# Residuals vs Leverage plot
```

```
plot(model_preferred,
     which = 5,
     pch = 21,
     bg = "#b3cccc",
     col = "#476b6b",
     cex = 0.8,
     lwd = 2)


par(mfrow = c(1, 1))




# --------------------------- END OF SCRIPT ---------------------------------------------
```